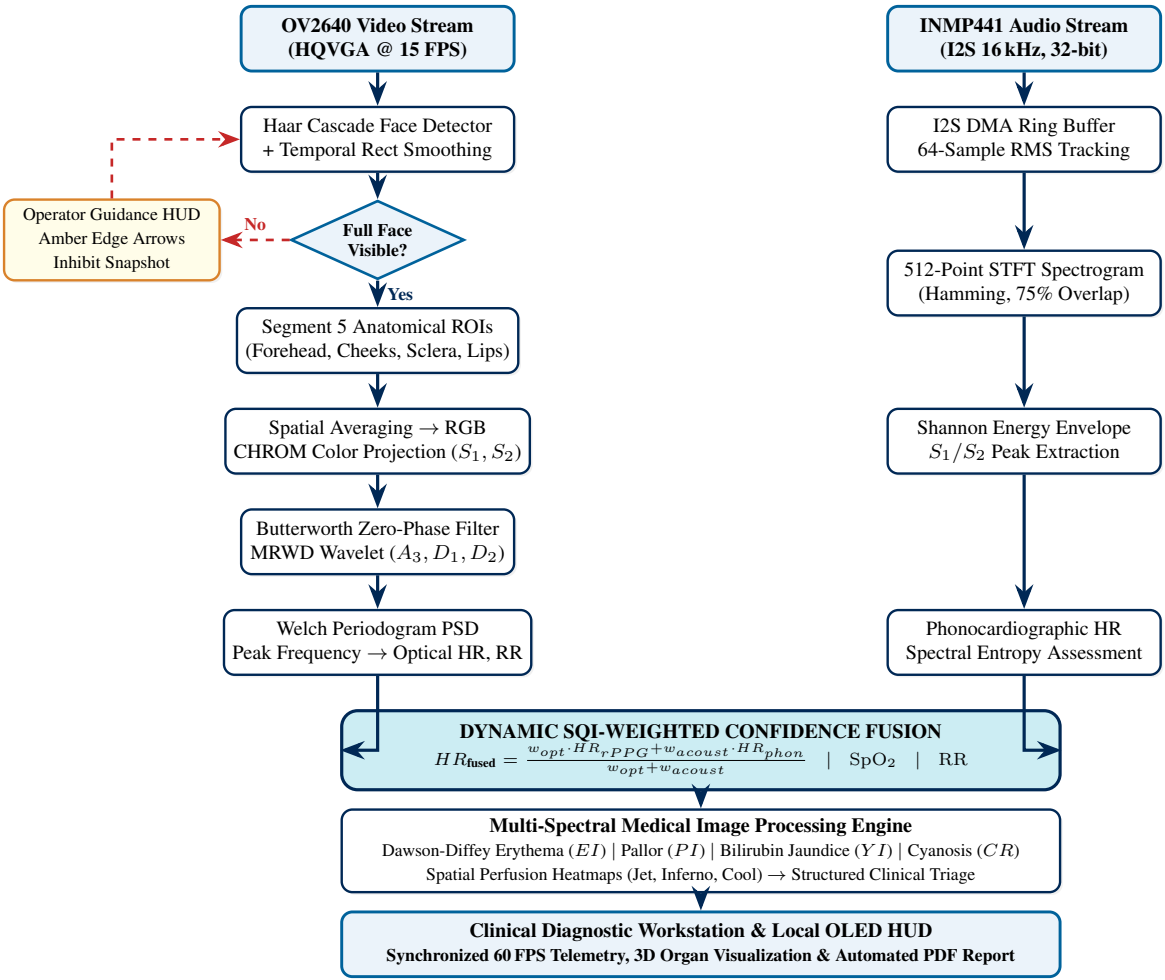

OV2640 Video Stream
(HQVGA @ 15 FPS)
Haar Cascade Face Detector
+ Temporal Rect Smoothing
Operator Guidance HUD
Amber Edge Arrows
Inhibit Snapshot
No
Full Face
Visible?
Yes
Segment 5 Anatomical ROIs
(Forehead, Cheeks, Sclera, Lips)
Spatial Averaging → RGB
CHROM Color Projection (S_1, S_2)
Butterworth Zero-Phase Filter
MRWD Wavelet (A_3, D_1, D_2)
Welch Periodogram PSD
Peak Frequency → Optical HR, RR
INMP441 Audio Stream
(I2S 16 kHz, 32-bit)
I2S DMA Ring Buffer
64-Sample RMS Tracking
512-Point STFT Spectrogram
(Hamming, 75% Overlap)
Shannon Energy Envelope
S_1/S_2 Peak Extraction
Phonocardiographic HR
Spectral Entropy Assessment
DYNAMIC SQI-WEIGHTED CONFIDENCE FUSION
$HR_{\text{fused}} = \frac{w_{opt} \cdot HR_{rPPG} + w_{acoust} \cdot HR_{phon}}{w_{opt} + w_{acoust}}$ | SpO_2 | RR
Multi-Spectral Medical Image Processing Engine
Dawson-Diffey Erythema (EI) | Pallor (PI) | Bilirubin Jaundice (YI) | Cyanosis (CR)
Spatial Perfusion Heatmaps (Jet, Inferno, Cool) → Structured Clinical Triage
Clinical Diagnostic Workstation & Local OLED HUD
Synchronized 60 FPS Telemetry, 3D Organ Visualization & Automated PDF Report